

(1001CJA100120006)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

09-08-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	B,C	A,B,C,D	A,B,D	A,C,D	A,B	A	B	B	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.39	28.28	0.80	7.50	4.00	3.46	0.40	7.50		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,B,C,D	A,C	B,D	A,B,C	B,C,D	B	B	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	300.00	5.00	2.00	6.00	12.80	40.00	2.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,C,D	B,C	A,B,C,D	A,B	B,D	A	C	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.66 or 0.67	3.33 or 3.34	2.00	9.00	9.50	0.00	3.00	0.60		

(1001CJA100120007)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

UNIT TEST

09-08-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	C	C	C	B	C	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	B	B	D	B	B	A	B	D
SECTION-II	Q.	1	2	3	4	5					
	A.	5.00	4.00	5.00	5.00	6.00					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	D	D	D	D	D	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	B	C	D	C	B	C	D	D
SECTION-II	Q.	1	2	3	4	5					
	A.	53.33	66.66 OR 66.67	6.00	18.00	7.00					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	D	B	B	C	A	A	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	D	A	A	A	B	A	C	D
SECTION-II	Q.	1	2	3	4	5					
	A.	2.00	160.00	1.50	1.25	7.82 or 7.83					

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A,C)

Sol. $2mg - T = 2ma$ $a = \frac{g}{3}$

$$T - mg = ma$$

Since acceleration is in opposite direction hence

$$a_{\text{rel}} = \frac{2g}{3}$$

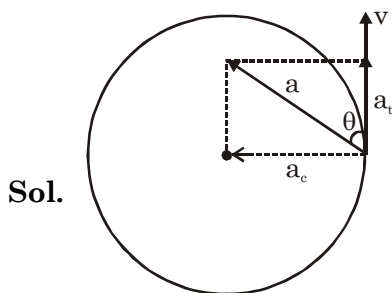
Work by tension of a system = 0

Heavy mass moving down, Hence G.P.E decreases.

2. Ans. (B,C)

Sol. In uniform circular motion, directions of velocity & acceleration changes.

3. Ans. (A,B,C,D)



$$a_t = a \cos \theta$$

$$a_c = a \sin \theta$$

$$\& a_c = \frac{v^2}{R} = \frac{u^2}{2} = 8 \text{ m/s}^2$$

$$\text{so } 8 = a \sin 30^\circ \Rightarrow a = 16 \text{ m/s}^2$$

$$\& a_t = 16 \cos 30^\circ = 8\sqrt{3} \text{ m/s}^2$$

$$\vec{a} \cdot \vec{v} = av \cos \theta = 16 \times 4 \times \cos 30^\circ = 32\sqrt{3}$$

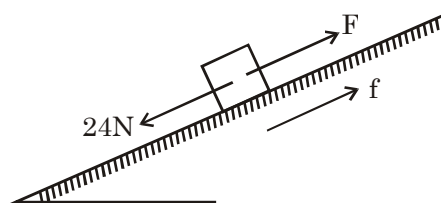
$$|\vec{a} \times \vec{v}| = av \sin \theta = 16 \times 4 \times \sin 30^\circ = 32$$

4. Ans. (A,B,D)

Sol. $(f_s)_{\text{max}} = \mu N = \mu mg \cos \theta = 0.5 \times 4 \times 10 \times \frac{4}{5}$
 $= 16 \text{ N}$

$$mg \sin \theta = 4 \times 10 \times \frac{3}{5} = 24 \text{ N}$$

As block is at rest $F + f = 24$



(A) $24 + f = 24 \Rightarrow f = 0$

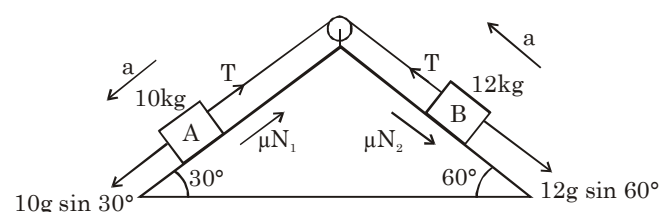
(B) $10 + f = 24 \Rightarrow f = 14 \text{ N}$

(C) $16 + f = 24 \Rightarrow f = 8 \text{ N}$

(D) $12 + f = 24 \Rightarrow f = 12 \text{ N}$

5. Ans. (A,C,D)

Sol.



A : $10g \sin 30^\circ - T - \mu N_1 = 10a$

& $N_1 = 10g \cos 30^\circ$

B : $T - \mu N_2 - 12g \sin 60^\circ = 12a$

& $N_2 = 12g \cos 60^\circ$

Solving we get $a = \left(\frac{6.8\sqrt{3} - 5}{22} \right) g$

$T = 55.79 \text{ N}$

$f_A = \mu N_1 = 35.92 \text{ N}$

6. Ans. (A, B)

Sol. (A) $\Delta V = 2V \sin \frac{\theta}{2}$

(B) $a_c = \frac{v^2}{R}$

(C) $\langle a \rangle = \frac{\Delta v}{\Delta t}$ where $\Delta t = \frac{2\pi R}{v} \times \frac{1}{3}$

7. Ans. (A)

Sol. (P) $(f_s)_{\max} = \mu_s N = 0.6 (2g) = 12 \text{ N}$

$f_k = \mu_k N = 0.5 (2g) = 10 \text{ N}$

block won't move, $a = 0$, $f = 10 \text{ N}$ so 2,4

(Q) $32 > (f_s)_{\max}$ so block will slip.

$[(f_s)_{\max} = 0.6 (4g) = 24 \text{ N}]$

$\& f_k = 0.5 (4g) = 20 \text{ N}]$

so f_k will act $\Rightarrow$ (1) so (A)

slip. $[(f_s)_{\max} = 0.6 (4g) = 24 \text{ N}]$

$\& f_k = 0.5 (4g) = 20 \text{ N}]$

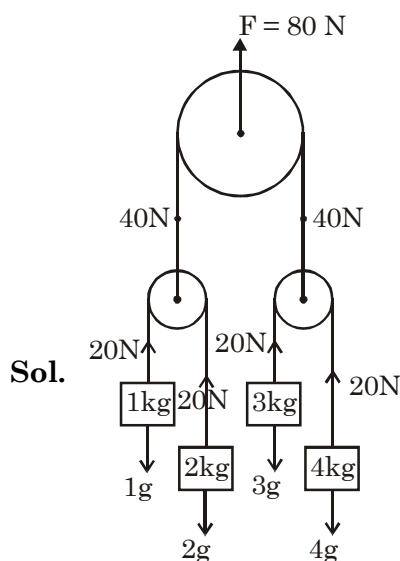
so f_k will act $\Rightarrow$ (1) so (A)

8. Ans. (B)

Sol. No slipping anywhere.

Net force is centripetal as $v = \text{constant}$

9. Ans. (B)



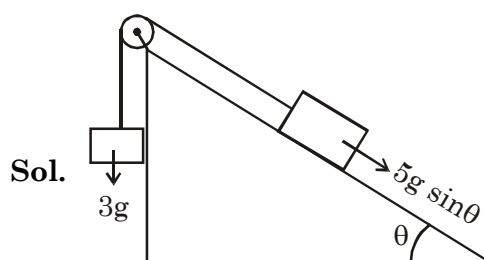
$a(1\text{kg}) = \frac{20 - 1g}{1} = 10 \text{ m/s}^2 (\uparrow)$

$a(2\text{kg}) = 0$

$a(3\text{kg}) = \frac{3g - 20}{3} = \frac{10}{3} \text{ m/s}^2 (\downarrow)$

$a(4\text{kg}) = \frac{4g - 20}{4} = 5 \text{ m/s}^2 (\downarrow)$

10. Ans. (D)



(A) $5g \sin \theta > 3g$

$\sin \theta > \frac{3}{5} \Rightarrow \theta > 37^\circ$

(B) $\frac{g}{8} = \frac{5g \sin \theta - 3g}{8}$

$5g \sin \theta = 4g$

$\theta = 53^\circ$

(C) $\frac{3g - 5g \sin \theta}{8} = \frac{g}{16}$

$3g - 5g \sin \theta = \frac{g}{2}$

$5g \sin \theta = \frac{5g}{2} \Rightarrow \theta = 30^\circ$

(D) $N = mg \cos \theta \quad N_{\min} \Rightarrow 0$

$N_{\max} \Rightarrow 50$

SECTION-II

1. Ans. 0.39

Sol. $(v_b)_x = 4 - 5 \sin 37^\circ = 1 \text{ m/s}$

$(v_b)_y = 5 \cos 37^\circ = 4 \text{ m/s}$

time to cross river = $\frac{s_y}{(v_b)_y} = \frac{1560}{4} = 390 \text{ sec}$

drift $s_x = (v_b)_x t = 1 \times 390 \text{ m} = 0.39 \text{ km}$

2. Ans. 28.28

Sol. $t = \sqrt{2h/g}$ & $R = v_x t$

3. Ans. 0.80

Sol. $a = \frac{F - F \cos 37^\circ}{2}$

4. Ans. 7.50

Sol. Apply $\Sigma \vec{T} \cdot \vec{a} = 0$

5. Ans. 4.00

Sol. Minimum radius of curvature will be at highest point.

Therefore

$$\frac{u^2}{g \cos \theta} = 8 \frac{u^2 \cos^2 \theta}{g}$$

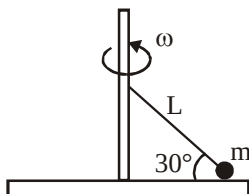
$$\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

6. Ans. 3.46

Sol. $T = 12g$ & $T = 10(1)\omega^2$

7. Ans. 0.40

Sol.



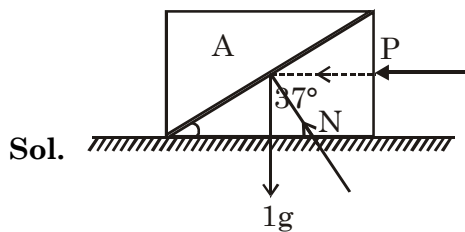
$$T \cos 30^\circ = m(L \cos 30^\circ) \omega^2 \quad \dots (i)$$

$$T \sin 30^\circ + N = mg$$

$$N = mg - T \sin 30^\circ \quad \dots (ii)$$

Putting value of T from equation in (ii) we get N.

8. Ans. 7.50



Sol.

$$N \cos 37^\circ = 1g \quad \dots (1)$$

$$N \sin 37^\circ = 1a \quad \dots (2)$$

$$\frac{a}{g} = \tan 37^\circ$$

$$a = 10 \times \frac{3}{4} = 7.5 \text{ m/s}^2$$

$$P = 2a$$

$$P = 15\text{N}$$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**

1. Ans.(A,B,C)
2. Ans.(A,B,C,D)
3. Ans.(A,C)
4. Ans.(B,D)

$$(A) \text{ No. of atoms in } N_2O = 3 \times \frac{10}{44} N_A$$

$$\text{No. of . of atoms in } CO_2 = 3 \times \frac{10}{44} N_A$$

$$(B) \text{ No. of atoms in } CO = 2 \times \frac{14}{28} N_A = 1N_A$$

$$\text{No. of atoms in } C_2H_4 = 6 \times \frac{14}{28} N_A = 3N_A$$

$$(C) \text{ No. of atoms in } SO_2 = 3 \times 4 N_A = 12N_A$$

$$\text{No. of atoms in } SO_3 = 4 \times 3 N_A = 12N_A$$

$$(D) \text{ No. of molecule in } O_2 = \frac{W}{32} \cdot N_A$$

$$\text{No. of molecule in } O_3 = \frac{W}{48} \cdot N_A$$

5. Ans.(A,B,C)

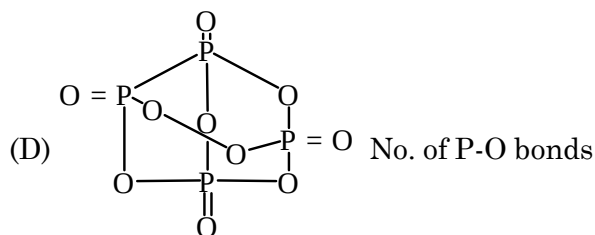
$$\text{No. of moles of } P_4O_6 = \frac{110}{220} = \frac{1}{2} \text{ moles.}$$

$$(A) \text{ No. of 'O' atoms} = \frac{1}{2} \times 6N_A = 3N_A$$

$$(B) \text{ No. of total atoms} = 10 \times \frac{1}{2} \times N_A = 5N_A$$

$$(C) \text{ Moles of 'P' atoms} = 4 \times \frac{1}{2} = 2 \text{ moles}$$

$$\text{weight of 'P' atoms} = 2 \times 31 = 62 \text{ gm}$$



$$= 16 \times \frac{1}{2} N_A = 8N_A$$

6. Ans.(B,C,D)

$$(A) \text{ Moles of 'O' atom} = 2 \times 0.05 + 2 \times 0.05$$

$$= 0.1 + 0.1 = 0.2 \text{ mole}$$

$$(B) \text{ Mass of 'O' atoms} = 0.2 + 16 = 3.2 \text{ gm}$$

$$(C) \text{ Total moles of atoms in } CO_2 = 3 \times 0.05$$

$$= 0.15$$

$$\text{Total moles of atoms in } O_2 = 0.05 \times 2 = 0.1$$

$$\text{Total moles of all atoms in container} = 0.15$$

$$+ 0.1 = 0.25$$

$$(D) \text{ Moles of carbon} = 0.05 \text{ mole}$$

$$W_C = 0.05 \times 12 = 0.6 \text{ gm}$$

$$\text{moles of carbon} = \frac{3.011 \times 10^{22}}{6.022 \times 10^{23}} = \frac{1}{2} \times 10^{-1}$$

$$W_C = \frac{1}{20} \times 12 = 0.6 \text{ gm}$$

7. Ans.(B)

$$(P) SO_3 \Rightarrow \text{V.D.} = \frac{MW}{2} = \frac{80}{2} = 40$$

$$\text{No. of electrons in one mole} = 40N_A$$

$$(Q) CO \Rightarrow \% \text{ by mass of C} = \frac{12}{28} \times 100$$

$$= \frac{300}{7} \%$$

$$(R) C_6H_6 \Rightarrow \text{Empirical formula} = C - H$$

$$\% \text{ by mole of C} = \frac{6}{12} \times 100 = 50\%$$

(S) $C_2H_2 \Rightarrow$ Empirical formula = C – H

$$\% \text{ by mole of C} = \frac{2}{4} \times 100 = 50\%$$

8. **Ans.(B)**

9. **Ans.(C)**

10. **Ans.(A)**

SECTION-II

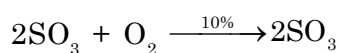
1. **Ans.(3.00)**

2. **Ans.(300.00)**

3. **Ans.(5.00)**

4. **Ans.(2.00)**

5. **Ans.(6.00)**



$$t=0 \quad \frac{50}{64} \quad \frac{13}{32}$$

$$= 0.78 \quad = 0.375 \text{ (L.R.)}$$

$$t=t \quad 0$$

$$= 2 \times \frac{10}{100} \times 0.375$$

$$= 0.075 \text{ mole}$$

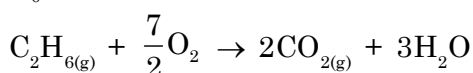
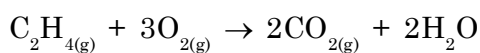
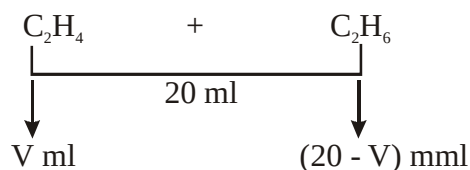
$$w = 6 \text{ gm.}$$

6. **Ans.(12.80)**

$$M_{\text{avg}} = \frac{X_1 M_1 + X_2 M_2 + \dots}{X_1 + X_2 + \dots}$$

$$= \frac{50(12) + 20(13) + 30(14)}{100} = 12.80$$

7. **Ans.(40.00)**



$$\text{total } CO_2 \text{ produced} = 2V + 2(20 - V) = 40 \text{ ml}$$

8. **Ans.(2.00)**

No. of moles of C-atom required for a line

$$= \frac{2.4 \times 10^{-6}}{12}$$

$$= 0.2 \times 10^{-6} \text{ moles}$$

So, required no. of moles of C-atoms for

$$10^7 \text{ lines} = 0.2 \times 10^{-6} \times 10^7 = 2$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (B,C,D)**

Sol. (A) $3 < 4$ & $11 > 9$

$$\log_2 3 < 2 \text{ \& } \log_3 11 > 2$$

(B) $\log_2 3 \log_3 4 \log_4 5 \dots \log_{1023} 1024$
 $= \log_2 1024 = 10$

(C) $\log_8 \left(\frac{8}{x^2} \right) = 3(\log_8 x)^2$
 $1 - \log_8(x^2) = 3(\log_8 x)^2$
 $1 - 2t = 3t^2$ where $t = \log_8 x$
 $3t^2 + 2t - 1 = 0$
 $t = \frac{1}{3}, -1$

$$\log_8 x = \frac{1}{3}, -1 \Rightarrow x = 2, \frac{1}{8}$$

(D) $\log_8 1 + \log_8 2 + \log_8 3 = \log_8(6)$
 $\log_8(6) = \log_8(1 + 2 + 3)$

2. **Ans. (A,C,D)**

Sol. $a^{\log_5 7} = 25$

$$a^{(\log_5 7)^2} = (a^{\log_5 7})^{\log_5 7} = (25)^{\log_5 7}$$

$$N^2 = (7)^{\log_5 25} = (7)^2$$

$$N = 7$$

3. **Ans. (B,C)**

Sol. Characteristic of $\log_5 m = 3$
 i.e. $m = [5^2, 5^3]$ i.e. $[25, 125]$

$$\text{So } \log_{25}(125) = \frac{3}{2}$$

4. **Ans. (A,B,C,D)**

Sol. $T_n = (an^2 + bn + c) \cdot 2^{(an^2 + bn)}$
 $\Rightarrow T_1 = (a + b + c)2^{a+b} = 1$
 $T_2 = (4a + 2b + c) \cdot 2^{4a+2b} = 2 \times 2^1$
 $T_3 = (9a + 3b + c) \cdot 2^{9a+3b} = 4 \times 2^3$

$$\Rightarrow c = 1, a = \frac{1}{2}, b = \frac{-1}{2}$$

Alternately the problem can also be solved by using the method for A.G.P. twice.

5. **Ans. (A,B)**

Sol. $\left(\frac{x-1}{x} \right) + \left(\frac{x-2}{x} \right) + \dots + \left(\frac{x-(x-1)}{x} \right) = 3$
 $\left(1 - \frac{1}{x} \right) + \left(1 - \frac{2}{x} \right) + \dots + \left(1 - \frac{x-1}{x} \right) = 3$
 $= (x-1) \cdot 1 - \left(\frac{1}{x} \right) (1 + 2 + \dots + (x-1))$
 $= (x-1) - \frac{1}{x} \frac{(x-1)x}{2} = 3$
 $= \frac{x-1}{2} = 3 \Rightarrow x = 7$

6. **Ans. (B,D)**

Sol. $T_r = \frac{r+2}{r(r+1)2^{r+1}} = \frac{1}{r \cdot 2^r} - \frac{1}{(r+1) \cdot 2^{r+1}}$
 $S_{15} = \left(\frac{1}{1 \cdot 2^1} - \frac{1}{2 \cdot 2^2} \right) + \left(\frac{1}{2 \cdot 2^2} - \frac{1}{3 \cdot 2^3} \right) + \dots + \left(\frac{1}{15 \cdot 2^{15}} - \frac{1}{16 \cdot 2^{16}} \right)$
 $\Rightarrow S_{15} = \frac{1}{2} - \frac{1}{2^{20}}$
 $\Rightarrow 2S_{15} = 1 - \frac{1}{2^{19}} \Rightarrow 1 - 2S_{15} = 2^{-19}$
 $\Rightarrow \log_2(1 - 2S_{15}) = -19$

7. **Ans. (A)**

Sol. (P) $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}} = y$
 $y^2 = 6 + y \Rightarrow y = 3, -2$ (Reject)
 $\log_2(\log_9 3) = \log_2 \left(\frac{1}{2} \right) = -1$

$$\text{(Q)} \log_{\sqrt{\frac{15}{8}}} \left(\frac{\log 3^{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}}}{\log \sqrt{3}} \right)$$

$$\log_{\sqrt{\frac{15}{8}}} \left(\frac{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}}{\frac{1}{2}} \right)$$

$$= \log_{\sqrt{\frac{15}{8}}} \left(\frac{15}{8} \right) = 2$$

(R) Do Yourself

(S) $\log_{\pi} 3 + \log_{\pi} 4 = \log_{\pi} 12$
 $\pi^2 < 12 < \pi^3$
 $2 < \log_{\pi} 12 < 3$

8. Ans. (C)

Sol. (P) $\cos 6x + \tan^2 x + \cos 6x \cdot \tan^2 x = 1$
 $\Rightarrow (1 + \tan^2 x)(1 + \cos 6x) = 2$
 $\Rightarrow \sec^2 x \cdot (1 + \cos 6x) = 2$
 $\Rightarrow 1 + \cos 6x = 2 \cos^2 x$
 $\Rightarrow \cos 6x = \cos 2x$
 $\Rightarrow \cos 6x - \cos 2x = 0$
 $\Rightarrow 2 \sin 4x \sin 2x = 0$

$$x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi$$

but $\frac{\pi}{2}, \frac{3\pi}{2}$ rejected (because $\tan x$ not defined)

so total solutions 7

(Q) $\tan x + \cot x = 2 \operatorname{cosec} x$

$$\Rightarrow \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{2}{\sin x}$$

$$\Rightarrow \sin^2 x + \cos^2 x = 2 \cos x$$

$$\Rightarrow \cos x = \frac{1}{2}$$

$$\Rightarrow 4 \text{ solutions in } [-2\pi, 2\pi]$$

(R) $6 \cos 2\theta + 2 \cos^2 \frac{\theta}{2} + 2 \sin^2 \theta = 0$

$$\Rightarrow 6 \cos 2\theta + 1 + \cos \theta + 2 \sin^2 \theta = 0$$

$$\Rightarrow 6[2 \cos^2 \theta - 1] + 1 + \cos \theta + 2[1 - \cos^2 \theta] = 0$$

$$\Rightarrow 10 \cos^2 \theta + \cos \theta - 3 = 0$$

$$\Rightarrow 10 \cos^2 \theta + 6 \cos \theta - 5 \cos \theta - 3 = 0$$

$$\Rightarrow 2 \cos \theta [5 \cos \theta + 3] - [5 \cos \theta + 3] = 0$$

$$\Rightarrow (2 \cos \theta - 1)(5 \cos \theta + 3) = 0$$

$$\Rightarrow \cos \theta = \frac{1}{2} \text{ or } \frac{-3}{5}$$

Total 4 solutions in $[-\pi, \pi]$

(S) $\sin^3 x \cos x + \sin^2 x \cos^2 x + \sin x \cos^3 x = 1$
 $\sin x \cos x [\sin^2 x + \cos^2 x] + \sin^2 x \cos^2 x = 1$

$$\Rightarrow \frac{\sin 2x}{2} + \frac{\sin^2 2x}{4} = 1$$

$$\Rightarrow \sin^2 2x + 2 \sin 2x - 4 = 0$$

$$\Rightarrow \sin 2x = \frac{-2 \pm \sqrt{4 + 16}}{2}$$

$$= -1 \pm \sqrt{5} \notin [-1, 1] \quad \text{both rejected}$$

9. Ans. (D)

Sol. (P) $3[\sin x + \cos x] - 2[\sin^3 x + \cos^3 x] = 8$
 $3[\sin x + \cos x] - 2[(\sin x + \cos x)^3 - 3 \sin x \cos x (\sin x + \cos x)] = 8 \quad \dots(1)$

We know

$$(\sin x + \cos x)^2 = 1 + 2 \sin x \cos x$$

$$\Rightarrow \sin x \cos x = \frac{(\sin x + \cos x)^2 - 1}{2}$$

Now put $\sin x + \cos x = t$ in equation (1)

$$3t - 2 \left(t^3 - 3 \left(\frac{t^2 - 1}{2} \right) t \right) = 8$$

Now Do your self

(Q) $\sin^4 x - \cos^2 x \sin x + 2 \sin^2 x + \sin x = 0$
 $\Rightarrow \sin^4 x + 2 \sin^2 x + \sin x [1 - \cos^2 x] = 0$
 $\Rightarrow \sin^4 x + 2 \sin^2 x + \sin^3 x = 0$
 $\Rightarrow \sin^2 x [\sin^2 x + 2 + \sin x] = 0$
 $\Rightarrow \sin x = 0$

(R) $x^2 + 2x \sin(xy) + 1 = 0$
 $\Rightarrow x^2 + 1 = -2x \sin(xy)$

$$\Rightarrow x + \frac{1}{x} = -2 \sin(xy)$$

L.H.S. R.H.S.

$$\in (-\infty, -2] \cup [2, \infty) \in [-2, 2]$$

so equality can be established if

$$\text{L.H.S.} = \text{R.H.S.} = 2 \text{ or } -2$$

$$\text{If L.H.S.} = \text{R.H.S.} = 2$$

$$\Rightarrow x = 1 \text{ \& } -\sin y = 2$$

$$\Rightarrow \sin y = -1 \Rightarrow y = \frac{3\pi}{2}$$

similarly for $x = -1$

$$\Rightarrow \sin y = 1 \Rightarrow y = \frac{\pi}{2}$$

(S) $\sin 3\theta - \sin \theta = 2[2 \cos^2 \theta - 1]$
 $\Rightarrow 2 \cos 2\theta \sin \theta = 2 \cos 2\theta$
 $\Rightarrow \cos 2\theta [\sin \theta - 1] = 0$
 $\Rightarrow \cos 2\theta = 0 \text{ or } \sin \theta = 1$

10. Ans. (C)

Sol. (P) $\sqrt{\sin^4 x + 4 \cos^2 x} - \sqrt{\cos^4 x + 4 \sin^2 x}$

$$= \sqrt{\sin^4 x + 4 - 4 \sin^2 x} - \sqrt{\cos^4 x + 4 - 4 \cos^2 x}$$

$$= (2 - \sin^2 x) - (2 - \cos^2 x)$$

$$= \cos^2 x - \sin^2 x = \cos 2x$$

(Q) $(\cos A - \cos B)^2 + (\sin A - \sin B)^2$

$$\Rightarrow 2 - 2[\cos A \cos B + \sin A \sin B]$$

$$\Rightarrow 2 - 2\cos(A - B)$$

$$\Rightarrow 2 \left[2 \sin^2 \left(\frac{A - B}{2} \right) \right] = 4 \sin^2 \left(\frac{A - B}{2} \right)$$

(R) $\sin A + \cos A \in [-\sqrt{2}, \sqrt{2}]$

$$\Rightarrow \sqrt{2} [\sin A + \cos A] \in [-2, 2]$$

(S) $\frac{\cos(\theta + \alpha) + \cos(\theta - \alpha)}{\cos(\theta + \alpha)\cos(\theta - \alpha)} = \frac{2}{\cos \theta}$

$$\Rightarrow \frac{2 \cos \theta \cos \alpha}{\cos^2 \theta - \sin^2 \alpha} = \frac{2}{\cos \theta}$$

$$\Rightarrow \cos^2 \theta [\cos \alpha - 1] = -\sin^2 \alpha$$

$$\Rightarrow \cos^2 \theta = \frac{\sin^2 \alpha}{1 - \cos \alpha}$$

$$= \frac{4 \sin^2 \frac{\alpha}{2} \cos^2 \frac{\alpha}{2}}{2 \sin^2 \frac{\alpha}{2}}$$

$$= 2 \cos^2 \frac{\alpha}{2}$$

$$\Rightarrow \cos \alpha = -\sin^2 \theta \leq 0$$

SECTION-II

1. Ans. 0.66 or 0.67

Sol. $2 \log \sqrt{6 - 2x} = \log(x - 1) + \log x$

$$\log(6 - 2x) = \log(x^2 - x)$$

$$6 - 2x = x^2 - x \Rightarrow x^2 + x - 6 = 0$$

$$x = 2, x = -3 \text{ Reject}$$

2. Ans. 3.33 or 3.34

Sol. Do Yourself.

3. Ans. 2.00

Sol. Do Yourself.

4. Ans. 9.00

Sol. $x^3 - 5x^2 + 3 = 0 = (x - \alpha)(x - \beta)(x - \gamma)$
 $x = 2$

$$-9 = (2 - \alpha)(2 - \beta)(2 - \gamma)$$

$$(\alpha - 2)(\beta - 2)(\gamma - 2) = 9$$

5. Ans. 9.50

Sol. $p = \frac{1}{1 - \frac{1}{2}} = 2$

$$q = 19$$

6. Ans. 0.00

Sol. $8x^{100} - 20x^4 + 3x^2 = 91$

$$\Rightarrow x^2[8x^{98} - 20x^2 + 3] = (13) \times (7) \times (\pm 1)^2$$

If $x \in I$ but R.H.S. is

L.H.S. must be divisible by a

divisible by a perfect square

(integer)² which is $(\pm 1)^2$

so check the equation only for $x = \pm 1$

which is rejected

7. Ans. 3.00

Sol. $\alpha < 1 < \beta$

$$\Rightarrow 2^\alpha < 2 < 2^\beta$$

Now put $2^x = t$

$$\Rightarrow t^2 - 2(k - 1)t + k = 0$$

where one root is less than 2 & other root greater than 2

$$\Rightarrow f(2) < 0 \Rightarrow 4 - 2(k - 1) \times 2 + k < 0$$

$$\Rightarrow -3k + 8 < 0$$

$$\Rightarrow k > \frac{8}{3}$$

so minimum possible integral value of k is 3

8. Ans. 0.60

Sol. We know

$$\frac{\sec 8A - 1}{\sec 4A - 1} = \frac{\tan 8A}{\tan 2A}$$

using componendo & dividendo

$$\frac{\sec 8A + \sec 4A - 2}{\sec 8A - \sec 4A} = \frac{\tan 8A + \tan 2A}{\tan 8A - \tan 2A}$$

$$= \frac{\sin 10A}{\sin 6A}$$

$$m = 6 \text{ \& } n = 10 \Rightarrow \frac{m}{n} = 0.60$$

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A)****Sol.** $[F] = \text{MLT}^{-2}$

$$F = \frac{10\text{kg} \times 10\text{m}}{10 \times 10\text{s}^2} = \frac{1\text{kgm}}{\text{s}^2} = 1\text{N}$$

2. **Ans. (A)**

Sol. Net displacement = $\sqrt{3^2 + 4^2 + 12^2}$
 $= \sqrt{9 + 16 + 144} = \sqrt{169} = 13\text{m}$

3. **Ans. (A)****Sol.** $W = \vec{F} \cdot \vec{d}$

$$\vec{d} = 2\hat{i} + 3\hat{j} + 2\hat{k}$$

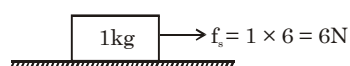
4. **Ans. (C)**

Sol. As no relative motion between blocks is the acceleration of both block will be same



$$\therefore a = \frac{18}{3} = 6\text{m/s}^2$$

$\therefore$ Force of friction between two blocks

5. **Ans. (C)**

Sol. As given the block moves with a constant velocity is friction force on block equals to $mg \sin \theta$. As block slides ie friction also equals to μN which is $\mu mg \cos \theta$.

6. **Ans. (C)****Sol.** $\Delta V = \text{area under a vs t graph}$ 7. **Ans. (B)****Sol.** $100 = Ma$

$$\& T = \frac{M}{2}a = 50\text{N}$$

8. **Ans. (C)****Sol.** $v_x = u_x + a_x t$

$$0 = 8 - 2t$$

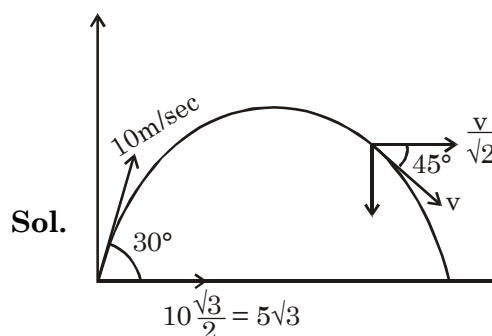
$$t = 4 \text{ sec.}$$

9. **Ans. (D)****Sol.** as acceleration $y = 0$

$$F_{\text{net}} = 0$$

10. **Ans. (B)****Sol.** $v_{\text{rel}} = 30 + 40 = 70 \text{ km/h}$

$$\text{time} = \frac{70\text{km}}{70\text{km/h}} = 1 \text{ hour.}$$

11. **Ans. (D)**

$$\frac{v}{\sqrt{2}} = 5\sqrt{3} \Rightarrow v = 5\sqrt{6}$$

$$\Rightarrow v^2 = 25 \times 6$$

$$= 150 \text{ m}^2/\text{s}^2$$

12. **Ans. (C)****Sol.** In frame of lift

$$u = 0, a = -10 \text{ m/s}^2, s = -20$$

$$-20 = \frac{1}{2} \times -10 \times t^2$$

$$t = 2\text{sec}$$

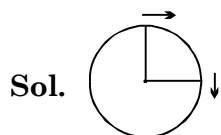
13. Ans. (B)

Sol. $a = \frac{20 + 3g - 2g}{2 + 3}$
 $= 6 \text{ m/s}^2$

14. Ans. (B)

Sol. Speed will be maximum for the time period upto which $\vec{a} \cdot \vec{v} > 0$

15. Ans. (D)



$$\vec{v}_1 = v\hat{i}$$

$$\vec{v}_2 = -v\hat{j}$$

$$\vec{v}_2 - \vec{v}_1 = -v(\hat{i} + \hat{j})$$

$$|\vec{v}_2 - \vec{v}_1| = \sqrt{2}v = \sqrt{2} \times \omega r$$

$$= \sqrt{2} \times \frac{2\pi}{60} \times 1 \text{ cm} = \frac{\pi\sqrt{2}}{30}$$

16. Ans. (B)

Sol. $\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{ rad}}{60 \text{ sec}} = \frac{3.14}{30} \text{ rad/sec.}$
 $= 0.105$

17. Ans. (B)

Sol. $KE = \frac{1}{2} \times 4 \times (2 \times 4)^2$
 $= 128 \text{ J}$

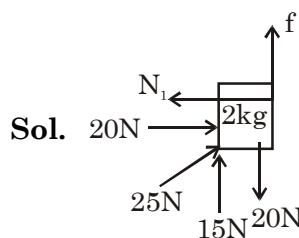
18. Ans. (A)

Sol. $a_c = \frac{v^2}{R} = \frac{(2 \times 4)^2}{4} = 16 \text{ m/s}^2$

19. Ans. (B)

Sol. $\Delta Q = \left(\frac{8 + 12}{2} \right) (4)$

20. Ans. (D)



$$f + 15 = 20$$

$$f = 5 \text{ N}$$

$$N_1 - 20 = 0$$

$$N_1 = 20$$

SECTION-II

1. Ans. 5.00

Sol. Net force = $Ma = 5 \times 1 = 5$

2. Ans. 4.00

Sol. $v_A \cos 37^\circ = v_B$

3. Ans. 5.00

Sol. $a = g \tan 53^\circ$

4. Ans. 5.00

Sol. $\mu mg = \frac{mv^2}{R}$

5. Ans. 6.00

Sol. $V_{\text{Net}} = \frac{40}{5} = 8 \text{ m/s}^2$

$$V_r = \sqrt{V_{b/r}^2 - V_{\text{Net}}^2}$$

$$= \sqrt{10^2 - 8^2} = 6 \text{ m/s}$$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans.(C)**2. **Ans.(B)**3. **Ans.(A)****Sol.** On the basis on bent's rule4. **Ans.(D)**5. **Ans.(D)**6. **Ans.(D)**7. **Ans.(D)**8. **Ans.(D)**9. **Ans.(D)**10. **Ans.(D)**11. **Ans.(D)**

P, T = same

 $V \propto n$

$$\Rightarrow \frac{V_{\text{CO}_2}}{V_{\text{He}}} = \frac{n_{\text{CO}_2}}{n_{\text{He}}}$$

$$\Rightarrow \frac{50}{V_{\text{He}}} = \frac{2}{0.5}$$

$$V_{\text{He}} = 12.5 \text{ L}$$

12. **Ans.(D)**

$$M_{\text{avg}} = \frac{\text{total mass}}{\text{total moles}}$$

$$= \frac{0.5 + 20 + 3.5}{\frac{0.5}{4} + \frac{20}{80} + \frac{3.5}{28}}$$

$$M_{\text{avg}} = 48 \text{ gm.}$$

13. **Ans.(B)**

$$M_w = 2 \times V.D. = 2 (13) = 26$$

$$\text{at STP} \Rightarrow V = 22.7 \times n$$

$$V = 18.16 \text{ L}$$

14. **Ans.(C)**

$$\begin{aligned} \text{(A) no. of C-atom} &= \frac{1 \times 1.6}{16} N_A \\ &= 0.1 N_A \text{ atoms.} \end{aligned}$$

$$\text{(B) In } C_3H_6 \Rightarrow \text{no. of C-atoms} = \frac{3 \times 8.4}{42} N_A$$

$$\text{(C) In } CH_3COOH$$

$$\Rightarrow \text{no. of C-atoms} = \frac{2 \times 24}{60} N_A = 0.8 N_A$$

$$\text{(D) No. of C-atoms} = \frac{1 \times 18.4}{46} N_A = 0.4 N_A$$

15. **Ans.(D)**

$$\text{No. of atoms in sulphur} = \frac{3.2}{32} N_A = 0.1 N_A$$

$$\text{(A)} = \frac{2.4}{24} N_A = 0.1 N_A$$

$$\text{(B)} \frac{1.2}{12} N_A = 0.1 N_A$$

$$\text{(C)} \frac{2.3}{23} N_A = 0.1 N_A$$

16. **Ans.(C)**

$$\text{no. of moles} = \frac{4.21 \times 10^{23}}{6.023 \times 10^{23}} = 0.699 \approx 0.7$$

$$\Rightarrow 0.7 = \frac{40.6}{\text{GMM}} \left[\frac{\text{Given amount}}{\text{GMM}} \right]$$

$$\text{GMM} = \frac{40.6}{0.7} = 58 \text{ gm} = C_4H_{10}$$

17. **Ans.(B)**

$$\begin{aligned} \text{No. of molecules} &= \frac{3.2}{64} N_A = 0.05 N_A \\ \text{molecules} \end{aligned}$$

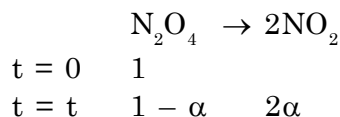
$$\text{(A) No. of } O_2 \text{ molecules} = \frac{8}{32} N_A = 0.25 N_A$$

$$\text{(B) No. of } H_2O \text{ molecule} = \frac{0.9}{18} N_A = 0.05 N_A$$

$$\text{(C) No. of He atom} = \frac{1}{4} N_A = 0.25 N_A$$

$$\text{(D) No. of H-atom} \left(\frac{1}{2} N_A \right) \times 2 = 1 N_A$$

18. Ans.(C)



$$M_{\text{avg}} = \frac{\text{Total mass}}{\text{Total moles}}$$

$$M_{\text{avg}} = \frac{1 \times 92}{1 - \alpha + 2\alpha} = \frac{92}{1 + \alpha} = \frac{92}{1.2} = 76.66$$

19. Ans.(D)

weight 1 atom of 'X' = Y amu

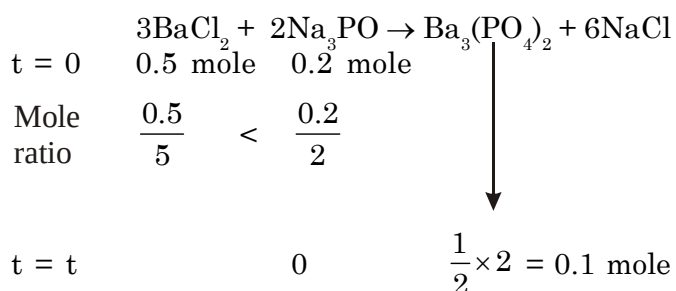
 weight of 1 molecule 'X₂' = 2 Y amu

 so, weight of N_A molecule = 2Y N_A amu =

$$2Y N_A \times \frac{1}{N_A} \text{ gm} = 2Y \text{ gm}$$

20. Ans.(D)

L.R.



SECTION-II

1. Ans. (53.33)

As moles 'O' atoms = 2 moles

 then mole of SO₃ = $\frac{2}{3}$ moles

$$W_{\text{SO}_3} = \frac{2}{3} \text{ moles} \times 80 = 53.33 \text{ or } 53.34$$

2. Ans. (66.66 or 66.67)

$$\text{No. of NH}_3 \text{ molecules} = \frac{3.4}{17} N_A$$

$$= 0.6 \times 6 \times 10^{23}$$

$$= 12 \times 10^{22} \text{ molecules.}$$

$$\text{No. of molecules left} = 8 \times 10^{22}$$

$$\% \text{ of molecules are left} = \frac{8 \times 10^{22}}{12 \times 10^{22}} \times 100$$

$$= 66.66 \text{ or } 66.67$$

3. Ans.(6.00)

4. Ans.(18.00)

 Sol. x = 6(XeO₃, SO₂, I₃⁺, IF₃, BrF₅, N(CH₃)₃)

 y = 3(XeO₃, SO₂, XeO₃F₂)

5. Ans.(7.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Sol. $(\sin\theta - 1)(\sin\theta - 2) = 0$

$$\sin\theta = 1 \quad \sin\theta = 2$$

$$\theta = \frac{\pi}{2} \text{ only}$$

2. **Ans. (A)**

Sol. $\frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = \frac{-b}{c}$

3. **Ans. (D)**

Sol. Do Yourself.

4. **Ans. (B)**

Sol. $a + b + c = -1$

$$\Rightarrow c = -2$$

$$\text{and } 8 + 4a + 2b + c = 0$$

$$\Rightarrow 4a + 2b = -6$$

$$\Rightarrow 2a + b = -3$$

$$\Rightarrow a = -4, b = 5$$

$$\text{hence } a = -4; b = 5; c = -2$$

5. **Ans. (B)**

Sol. Let $\tan \theta = x = \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots \infty}}} = \frac{1}{2 + x}$

$$x^2 + 2x - 1 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{8}}{2} = (\sqrt{2} - 1)$$

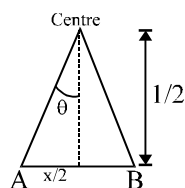
$$\therefore \tan \theta = \sqrt{2} - 1$$

$$\Rightarrow \theta = \frac{\pi}{8} \text{ or } \frac{9\pi}{8}$$

6. **Ans. (C)**

Sol. $\theta = 22.5^\circ$

$$\tan 22.5^\circ = \frac{x}{2} \cdot \frac{2}{1}$$



$$x = \tan 22.5^\circ = \sqrt{2} - 1$$

7. **Ans. (A)**

Sol. $S = 1 + 0$

$$(\because \sec \frac{\pi}{5} + \sec \frac{4\pi}{5} = 0 \text{ and } \sec \frac{2\pi}{5} + \sec \frac{3\pi}{5} = 0)$$

$$\text{now } P = \tan 20^\circ \cdot \tan 40^\circ \cdot \tan 80^\circ$$

$$\text{now } \sin 20^\circ \cdot \sin 40^\circ \cdot \sin 80^\circ = \frac{\sqrt{3}}{8}$$

$$\text{and } \cos 20^\circ \cdot \cos 40^\circ \cdot \cos 80^\circ = \frac{1}{8}$$

$$\text{dividing } \tan 20^\circ \cdot \tan 40^\circ \cdot \tan 80^\circ = \sqrt{3}$$

$$\therefore P = \sqrt{3}$$

$$\therefore S + P = 1 + \sqrt{3} = \left(\frac{\sqrt{3} + 1}{2\sqrt{2}} \right) (2\sqrt{2})$$

$$= 2\sqrt{2} \sin \frac{7\pi}{12} \quad \text{Hence proved.}$$

8. **Ans. (A)**

Sol. $x^{\frac{1}{2} \log \frac{(x^2 - x)}{(1/2) \log x}} = 3^{\log_3 2}$

$$\Rightarrow x^{\log_x (x^2 - x)} = 2$$

$$\Rightarrow x^2 - x = 2$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$\Rightarrow x = 2 \text{ or } x = -1 \quad (\text{rejected})$$

$$\therefore x = 2 \text{ Ans.}$$

9. **Ans. (D)**

Sol. $x^2y - 4xy - 5y = x^2 - 2x - 8$

$$(y - 1)x^2 + 2x(1 - 2y) + 8 - 5y = 0$$

$$\therefore x \in \mathbb{R} \text{ hence } D \geq 0$$

$$4(1 - 2y)^2 - 4(y - 1)(8 - 5y) \geq 0$$

$$(4y^2 - 4y + 1) - (13y - 8 - 5y^2) \geq 0$$

$$9y^2 - 17y + 9 \geq 0 \quad \dots(1)$$

$$\text{since coefficient of } y^2 > 0$$

$$\text{and } D = 289 - 324 < 0$$

$$\text{Hence (1) is always true}$$

$$\text{Hence range of } y \text{ is } (-\infty, \infty)$$

10. Ans. (B)

$$\text{Sol. } \cot^2 18^\circ = \frac{1 + \cos 36^\circ}{1 - \cos 36^\circ}$$

$$= \frac{1 + \frac{\sqrt{5}+1}{4}}{1 - \frac{\sqrt{5}+1}{4}} = \frac{(5+\sqrt{5})(3+\sqrt{5})}{(3-\sqrt{5})(3+\sqrt{5})}$$

$$= \frac{20+5\sqrt{5}+3\sqrt{5}}{9-5}$$

$$= \frac{20+8\sqrt{5}}{4} = 5+2\sqrt{5}$$

Hence if $\alpha = 5+2\sqrt{5}$

$$\beta = 5-2\sqrt{5}$$

$$\therefore \alpha + \beta = 10; \alpha\beta = 25 - 20 = 5$$

$\therefore$ quadratic equation is $x^2 - 10x + 5 = 0$ Ans.

11. Ans. (A)

Sol. Quadratic equation is

$$3x^2 - 8x + 1 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$\alpha + \beta = \frac{8}{3} \text{ and } \alpha\beta = \frac{1}{3}$$

to find the value of

$$\frac{1}{\alpha\beta + (\alpha + \beta) + 1} + \frac{1}{\alpha\beta - 2(\alpha + \beta) + 4} + \frac{1}{\alpha\beta - 3(\alpha + \beta) + 9}$$

$$= \frac{1}{4} - 1 + \frac{3}{4}$$

$$= -\frac{3}{4} + \frac{3}{4} = 0 \text{ Ans.}$$

12. Ans. (A)

Sol. $S_5 = 189$; $S_6 = 381$; $S_7 = 765$

$$t_6 = S_6 - S_5 = 381 - 189 = 192$$

$$t_7 = S_7 - S_6 = 765 - 381 = 384$$

$$\text{now common ratio} = \frac{t_7}{t_6} = \frac{384}{192} = 2 \text{ Ans.}$$

13. Ans. (D)

$$\text{Sol. } 2b = x + a \quad \dots(1)$$

$$b^2 = ay \quad \dots(2)$$

$$b = \frac{2az}{a+z} \quad \dots(3)$$

$$x = 2b - a; y = \frac{b^2}{a}$$

$$\text{and } \frac{2}{b} = \frac{1}{a} + \frac{1}{z}$$

$$\Rightarrow z = \frac{ab}{2a-b}$$

$$\therefore xyz = (2b-a) \frac{b^2}{a} \cdot \frac{ab}{2a-b}$$

$$= \frac{b^3(2b-a)}{2a-b} \text{ Ans.}$$

14. Ans. (A)

$$\text{Sol. } \frac{x^2 + 10x - 36}{x(x-3)^2} = \frac{a(x-3)^2 + b(x-3)x + cx}{x(x-3)^2}$$

$$\text{hence } x^2 + 10x - 36 = a(x-3)^2 + b(x-3)x + cx$$

$$\text{put } x = 0; -36 = 9a$$

$$\Rightarrow a = -4$$

$$x^2 + 10x - 36 = x^2(-4+b) + x(24-3b+c) + (-36)$$

comparing coefficients

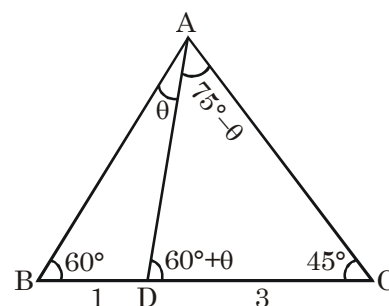
$$\text{also, } -4 + b = 1 \Rightarrow b = 5$$

$$24 - 15 + c = 10 \Rightarrow 9 + c = 10 \Rightarrow c = 1$$

$$a = -4; b = 5; c = 1 \quad \text{i.e. } a + b + c = 2 \text{ Ans.}$$

15. Ans. (A)

Sol.



$$\text{In } \triangle BAD \quad \frac{BD}{\sin \theta} = \frac{AD}{\sin 60^\circ}$$

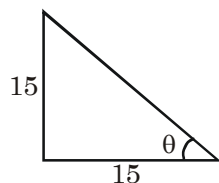
$$\text{In } \triangle CAD \quad \frac{CD}{\sin(75^\circ - \theta)} = \frac{AD}{\sin 45^\circ}$$

$$\Rightarrow \frac{BD}{\sin \theta} \sin 60^\circ = \frac{CD \sin 45^\circ}{\sin(75^\circ - \theta)}$$

$$\Rightarrow \frac{\sin \theta}{\sin(75^\circ - \theta)} = \frac{BD \sin 60^\circ}{CD \sin 45^\circ} = \frac{1}{\sqrt{6}}$$

16. Ans. (A)

Sol.



$$\tan \theta = \frac{15}{15}$$

$$\theta = 45^\circ$$

17. Ans. (B)

Sol. $\frac{y+z}{2} = \sqrt{yz} \Rightarrow 1-x \geq 2\sqrt{yz}$

Thus,

$$(1-x)(1-y)(1-z) \geq 2\sqrt{yz} \cdot 2\sqrt{zx} \cdot 2\sqrt{xy} = 8xyz$$

$$\Rightarrow \frac{xyz}{(1-x)(1-y)(1-z)} \leq \frac{1}{8}$$

18. Ans. (A)

Sol. By sine rule

19. Ans. (C)

Sol. $(x-1)(x+1) < 0$

$$x \in (-1, 1)$$

Integral value $x = 0$ only

20. Ans. (D)

Sol. $\frac{29}{12} = 2 + \frac{5}{12} = 2 + \frac{1}{\frac{12}{5}} = 2 + \frac{1}{2 + \frac{2}{5}}$

$$= 2 + \frac{1}{2 + \frac{1}{5/2}} = 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}$$

$$\Rightarrow a = 2, b = 2, c = 2, d = 2$$

$$a^3 + b^3 + c^3 + d^3 = 4 \times 8 = 32$$

SECTION-II

1. Ans. 2.00

Sol. Solving using the Identity

$$\tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta} \text{ where } \theta = \frac{\pi}{12}$$

$$\tan \frac{\pi}{24} = \frac{1 - \cos 15^\circ}{\sin 15^\circ}$$

$$= \frac{1 - \frac{\sqrt{6} + \sqrt{2}}{4}}{\frac{\sqrt{6} - \sqrt{2}}{4}}$$

$$= \frac{4 - (\sqrt{6} + \sqrt{2})}{\sqrt{6} - \sqrt{2}}$$

$$= \frac{[4 - (\sqrt{6} + \sqrt{2})][\sqrt{6} + \sqrt{2}]}{4}$$

$$= \frac{4(\sqrt{6} + \sqrt{2}) - (8 + 4\sqrt{3})}{4}$$

$$= (\sqrt{6} + \sqrt{2}) - (2 + \sqrt{3})$$

$$= (\sqrt{6} - \sqrt{3}) - (2 - \sqrt{2})$$

$$= \sqrt{3}(\sqrt{2} - 1) - \sqrt{2}(\sqrt{2} - 1)$$

$$= (\sqrt{3} - \sqrt{2})(\sqrt{2} - 1)$$

$$\text{hence } p = 3, q = 2; r = 2; s = 1$$

2. Ans. 160.00

Sol. Let H = set of persons who know Hindi.

K = set of persons who know Kannada.

$n(H \cap K)$ = the number of persons who know both Hindi and Kannada is 70.

$$n(H \cap K) = n(H) + n(K) - n(H \cup K)$$

$$= 150 + 80 - 70 = 160$$

3. Ans. 1.50

$$\begin{aligned}\text{Sol. LHS} &= \sin^2\theta + 1 - [\cos^2(120^\circ + \theta) - \sin^2(120^\circ - \theta)] \\ &= \sin^2\theta + 1 - [\cos 240^\circ - \cos 2\theta] \\ &= \sin^2\theta + 1 + \frac{1}{2} \cos 2\theta \\ &= \sin^2\theta + 1 + \frac{1}{2} (1 - 2 \sin^2\theta) = 3/2\end{aligned}$$

4. Ans. 1.25

$$\begin{aligned}\text{Sol. } \alpha + \frac{2\alpha}{3} &= \frac{10}{m} \\ \Rightarrow \frac{5\alpha}{3} &= \frac{10}{m} \\ \Rightarrow \alpha &= \frac{6}{m}; \text{ also } \alpha \cdot \frac{2\alpha}{3} = \frac{3}{m} \\ \Rightarrow 2\alpha^2 &= \frac{9}{m} \\ 2 \cdot \frac{36}{m^2} &= \frac{9}{m} \\ \Rightarrow \frac{24}{m} &= 3 \Rightarrow m = 8 \\ \therefore \text{sum} &= \frac{10}{m} = \frac{10}{8} = \frac{5}{4} \text{ Ans.}\end{aligned}$$

5. Ans. 7.82 or 7.83

Sol. Let $x = \cos \theta$ and $y = \sin \theta$

$$\begin{aligned}y = f(\theta) &= 7 \cos^2\theta + 4 \sin \theta \cos \theta + 3 \sin^2\theta \\ &= 3 + 2 \sin 2\theta + 2(1 + \cos 2\theta) \\ &= 5 + 2(\sin 2\theta + \cos 2\theta)\end{aligned}$$

$$\text{but } -\sqrt{2} \leq (\sin 2\theta + \cos 2\theta) \leq \sqrt{2}$$

$$\therefore y_{\max} = 5 + 2\sqrt{2} \text{ and } y_{\min} = 5 - 2\sqrt{2}$$